Stopping times for Brownian motion

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Definitions

Let $(\Omega, \mathcal{F}, F)$ be a filtered space.

Markov moment

Measurable function $\tau(\omega) : \Omega \to [0, \infty]$ is called Markov moment (with respect to filtration $F = (\mathcal{F}_t)_{t \geq 0}$) if $\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$ for every $t \geq 0$.

Stopping time

Stopping time is a Markov moment τ for which $\mathrm{P}(\tau(\omega) < \infty) = 1$.

Outline

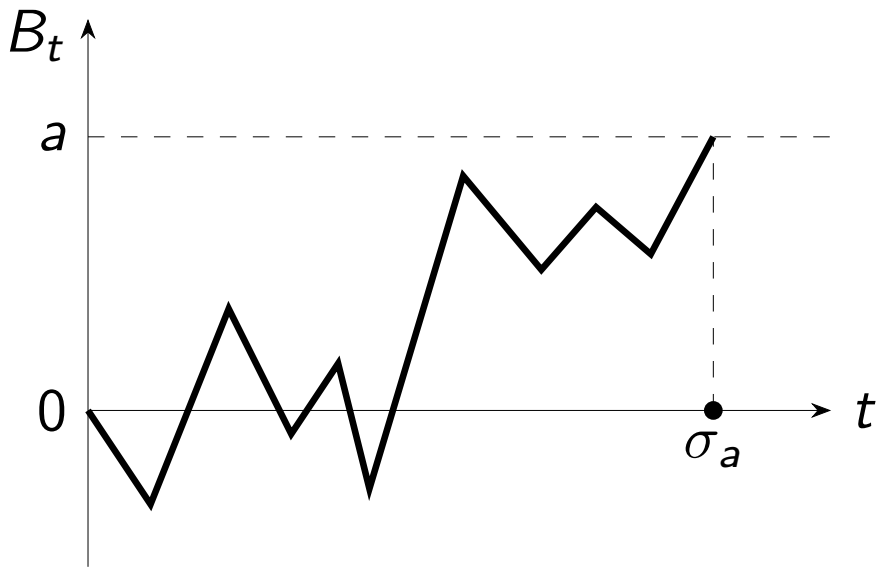
1 Hitting times
- Horizontal boundaries

2 Drawdowns

3 Hitting times
- Linear boundaries
- Non-linear boundaries

4 References

Properties of $\sigma_a = \inf\{t \geq 0 : B_t \geq a\}$, $a > 0$



Doob's optional sampling theorem

The theorem

Let $X = (X_n, \mathcal{F}_n)_{n\geq 0}$ be a martingale and τ, σ two finite stopping times such that $\mathrm{E}(X_\tau)$ and $\mathrm{E}(X_\sigma)$ are defined. If

$$\varliminf_{n\to\infty} \mathrm{E}[|X_n| I(\tau > m)] = 0, \text{ then}$$

$$\mathrm{E}(X_\tau|\mathcal{F}_\sigma) = X_{\tau\wedge\sigma} \quad (\mathrm{E}(X_\tau|\mathcal{F}_\sigma) = X_\sigma \ \{\tau \geq \sigma\}).$$

Corollary

If for any finite number T the condition $\mathrm{P}(\tau < T) = 1$ is satisfied then if X is a martingale

$$\mathrm{E}X_0 = \mathrm{E}X_\tau = \mathrm{E}X_T.$$

Passage to the limit under the expectation sign

Monotone convergence theorem

Let $\eta, \xi_1, \xi_2, \ldots$ be random variables.
a) If $\xi_n \geq \eta$ for all $n \geq 1$, $\mathrm{E}\eta > -\infty$ and $\xi_n \uparrow \xi$ ($\lim_{n\to\infty} \xi_n = \xi$ and $\xi_1 \leq \xi_2 \leq \ldots$),

$$\lim_{n\to\infty} \mathrm{E}\xi_n = \mathrm{E}\xi,$$

where $\mathrm{E}\xi_1 \leq \mathrm{E}\xi_2 \leq \ldots$. Can be written as $\mathrm{E}\xi_n \uparrow \mathrm{E}\xi$.
b) If $\xi_n \leq \eta$ for all $n \geq 1$, $\mathrm{E}\eta > -\infty$ and $\xi_n \downarrow \xi$ ($\lim_{n\to\infty} \xi_n = \xi$ and $\xi_1 \geq \xi_2 \geq \ldots$),

$$\lim_{n\to\infty} \mathrm{E}\xi_n = \mathrm{E}\xi,$$

where $\mathrm{E}\xi_1 \geq \mathrm{E}\xi_2 \geq \ldots$. Can be written as $\mathrm{E}\xi_n \downarrow \mathrm{E}\xi$.

Lebesgue dominated convergence theorem

Let the random variables be such that $|\xi_n| \leq \eta$, $\mathrm{E}\eta < \infty$ and $\xi_n \to \xi$. Then $\mathrm{E}|\eta| < \infty$ and

$$\mathrm{E}\xi_n \to \mathrm{E}\xi, \quad n \to \infty,$$

$$\mathrm{E}|\xi_n - \xi| \to 0, \quad n \to \infty.$$

Properties of $\sigma_a = \inf\{t \geq 0 : B_t \geq a\}$, $a > 0$

1. Apply optional sampling theorem to $M_{\sigma_a \wedge t} = e^{\theta B_{\sigma_a \wedge t} - \frac{\theta^2}{2}(\sigma_a \wedge t)}$ and get that
$$\mathrm{E}M_{\sigma_a \wedge t} = \mathrm{E}M_0 = 1.$$
2. Apply Lebesgue dominated convergence theorem to $\lim_{t\to\infty} \mathrm{E}M_{\sigma_a \wedge t}$ and obtain
$$1 = \lim_{t\to\infty} \mathrm{E}M_{t\wedge\sigma_a} = \mathrm{E} \lim_{t\to\infty} M_{t\wedge\sigma_a}.$$
3. Apply Strong law or large numbers for Brownian motion
$$\frac{B_t}{t} \to 0, \ t \to \infty \ (\mathrm{P}\text{-a.s.}).$$
4. Apply Lebesgue dominated convergence theorem for $\theta \downarrow 0$
$$1 = \lim_{\theta\to 0} \mathrm{E}M_{\sigma_a} I(\sigma_a < \infty) = \mathrm{E} \lim_{\theta\to 0} M_{\sigma_a} \mathrm{I}(\sigma_a < \infty) = \mathrm{P}(\sigma_a < \infty).$$
5. Obtain the Laplace transform for σ_a
$$\mathrm{E}e^{-\lambda\sigma_a} = e^{-\sqrt{2\lambda}a}.$$

Properties of $\sigma_a = \inf\{t \geq 0 : B_t \geq a\}$, $a > 0$

Using the fact that

$$\int_0^\infty e^{-\lambda t} \frac{a\phi_t(a)}{t} dt = e^{-a\sqrt{2\lambda}}, \text{where } \phi_t(a) = \frac{1}{\sqrt{2\pi t}} e^{\frac{-a^2}{2t}}$$

we get that pdf of σ_a is

$$p_{\sigma_a} = \frac{a\phi_t(a)}{t} = \frac{a}{\sqrt{2\pi t^3}} e^{\frac{-a^2}{2t}}.$$

Properties of $\sigma_a = \inf\{t \geq 0 : B_t \geq a\}$, $a > 0$

Using the fact that

$$\int_0^t \frac{a\phi_s(a)}{s} ds = 2\int_a^\infty \phi_t(x)dx$$

we get that cdf of σ_a is

$$\mathrm{P}(\sigma_a \leq t) = 2\mathrm{P}(B_t \geq a) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right).$$

Properties of $\sigma_a = \inf\{t \geq 0 : B_t \geq a\}$, $a > 0$

The **Laplace transform** of σ_a is

$$\mathrm{E}e^{-\lambda\sigma_a} = e^{-\sqrt{2\lambda}a}.$$

Probability density function of σ_a is

$$p_{\sigma_a} = \frac{a\phi_t(a)}{t} = \frac{a}{\sqrt{2\pi t^3}} e^{\frac{-a^2}{2t}}.$$

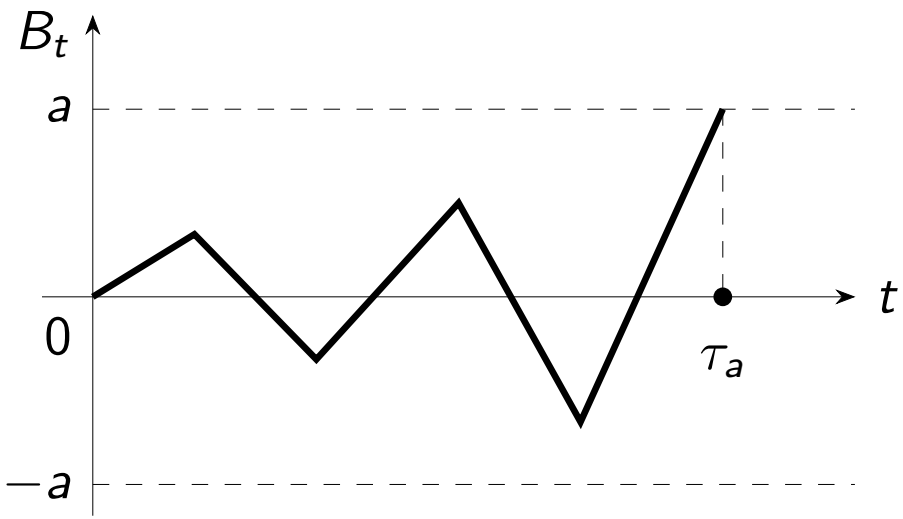
Cumulative density function of σ_a is

$$\mathrm{P}(\sigma_a \leq t) = 2\mathrm{P}(B_t \geq a) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right).$$

Expectation of σ_a is

$$\mathrm{E}\sigma_a = \infty.$$

Properties of $\tau_a = \inf\{t \geq 0 : |B_t| \geq a\}$, $a > 0$



Properties of $\tau_a = \inf\{t \geq 0 : |B_t| \geq a\}$, $a > 0$

1. Apply optional sampling theorem to $M_{\tau_a \wedge t} = e^{\theta B_{\tau_a \wedge t} - \frac{\theta^2}{2}(\tau_a \wedge t)}$ and get that
$$\mathrm{E}M_{\tau_a \wedge t} = \mathrm{E}M_0 = 1.$$
2. Apply Lebesgue dominated convergence theorem to $\lim_{t\to\infty} \mathrm{E}M_{\tau_a \wedge t}$ and obtain
$$1 = \lim_{t\to\infty} \mathrm{E}M_{t\wedge\tau_a} = \mathrm{E} \lim_{t\to\infty} M_{t\wedge\tau_a} = \mathrm{E}M_{\tau_a}.$$
3. Apply symmetry of Brownian motion $B_t \stackrel{\mathcal{D}}{=} -B_t$ to M_{τ_a}
$$\mathrm{E}M_{\tau_a} = \mathrm{E}M_{\tau_a} I(B_{\tau_a} = a) + \mathrm{E}M_{\tau_a} I(B_{\tau_a} = -a) = \frac{e^{\theta a} + e^{-\theta a}}{2} \mathrm{E}e^{-\frac{\theta^2}{2}\tau_a}$$
4. Obtain Laplace transform for τ_a
$$\mathrm{E}e^{-\lambda\tau_a} = \frac{1}{\mathrm{ch}(\sqrt{2\lambda}a)} = 2\sum_{n=0}^{\infty} (-1)^n e^{-a\sqrt{2\lambda}(2n+1)}.$$

Properties of $\tau_a = \inf\{t \geq 0 : |B_t| \geq a\}$, $a > 0$

Using the Laplace transform of σ_A

$$e^{-A\sqrt{2\lambda}} = \mathrm{E}e^{-\lambda\sigma_A} = \int_0^\infty e^{-\lambda t}\frac{A\phi_t(A)}{t}dt, \text{ where } A = a(2n+1)$$

we get that pdf of σ_a is

$$p_{\tau_a} = 2\sum_{n=0}^{\infty}(-1)^n\frac{A\phi_t(A)}{t} = 2\sum_{n=0}^{\infty}(-1)^n\frac{a(2n+1)}{t}\phi_t(a(2n+1)).$$

Properties of $\tau_a = \inf\{t \geq 0 : |B_t| \geq a\}$, $a > 0$

Expectation of τ_a is

$$\mathrm{E}\tau_a = a^2.$$

Laplace transform of τ_a

$$\mathrm{E}e^{-\lambda\tau_a} = \frac{1}{\mathrm{ch}(\sqrt{2\lambda}a)} = 2\sum_{n=0}^{\infty}(-1)^n e^{-a\sqrt{2\lambda}(2n+1)}.$$

Probability density function of τ_a is

$$p_{\tau_a} = 2\sum_{n=0}^{\infty}(-1)^n \frac{a(2n+1)}{t}\phi_t(a(2n+1)) = 2\sum_{n=0}^{\infty}(-1)^n p_{\sigma_{a(2n+1)}}(t).$$

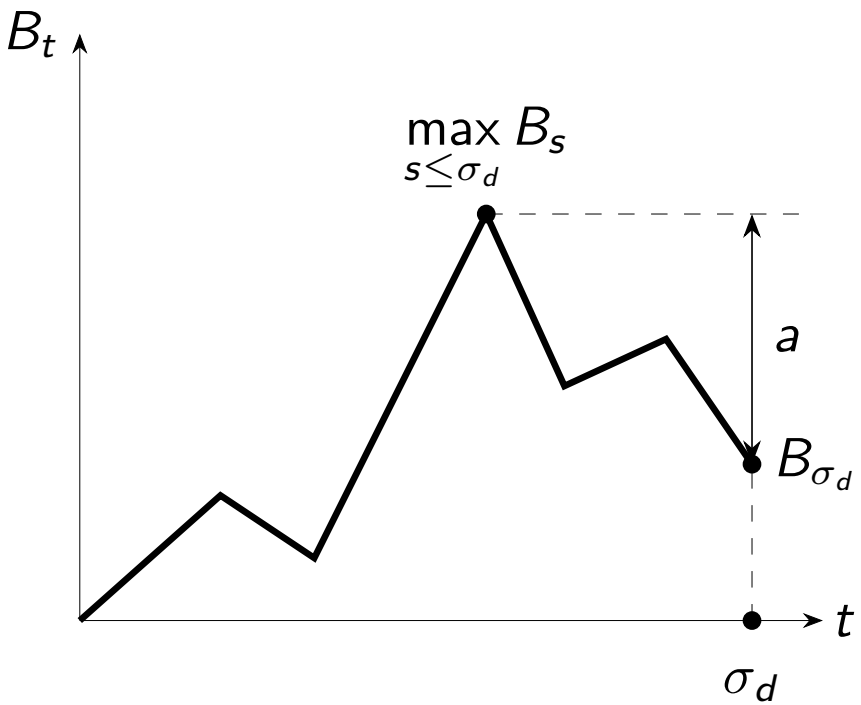
Cumulative density function of τ_a is

$$P(\tau_a \leq t) = \int_0^t p_{\tau_a}(s)ds = 4\sum_{n=0}^{\infty}(-1)^n \left[1 - \Phi\left(\frac{a(2n+1)}{\sqrt{t}}\right)\right].$$

Properties of $\tau_a = \inf\{t \geq 0 : |B_t| \geq a\}$, $a > 0$

1. Apply optional sampling theorem to $M_{\tau_a \wedge t} = B^2_{\tau_a \wedge t} - (\tau_a \wedge t)$ and get that
$$\mathrm{E}M_{\tau_a \wedge t} = \mathrm{E}M_0 = 0.$$
2. Apply Lebesgue dominated convergence theorem to $\lim_{t\to\infty} \mathrm{E}(t \wedge \tau_a)$ and obtain
$$\lim_{t\to\infty} \mathrm{E}(t \wedge \tau_a) = \mathrm{E} \lim_{t\to\infty} (t \wedge \tau_a) = \mathrm{E}\tau_a.$$
3. Apply Monotone convergence theorem to $\lim_{t\to\infty} \mathrm{E}B^2_{t\wedge\tau_a}$ and get
$$\lim_{t\to\infty} \mathrm{E}B^2_{t\wedge\tau_a} = \mathrm{E} \lim_{t\to\infty} B^2_{t\wedge\tau_a} = \mathrm{E}B^2_{\tau_a}.$$
4. Obtain the expectation of τ_a
$$\mathrm{E}\tau_a = \mathrm{E}B^2_{\tau_a}.$$

Properties of $\sigma_d = \inf\{t \geq 0 : \max_{s \leq t} B_s - B_t \geq a\}$, $a > 0$



Properties of $\sigma_d = \inf\{t \geq 0 : \max_{s \leq t} B_s - B_t \geq a\}$, $a > 0$

Levy's theorem

Let $B = (B_t)_{t \geq 0}$ is a standard Brownian motion. Then

$$(\sup B, \sup B - B) \stackrel{\mathcal{D}}{=} (L, |B|),$$

where L is local time of Brownian motion.

Then $\text{Law}(\sigma_d) = \text{Law}(\tau_a)$, and therefore σ_d has the same properties as τ_a.

Properties of $\sigma_d = \inf\{t \geq 0 : \max_{s \leq t} B_s - B_t \geq a\}$, $a > 0$

Expectation of σ_d is

$$\mathrm{E}\sigma_d = a^2.$$

Laplace transform of σ_d

$$\mathrm{E}e^{-\lambda\sigma_d} = 2\sum_{n=0}^{\infty}(-1)^n e^{-a\sqrt{2\lambda}(2n+1)}.$$

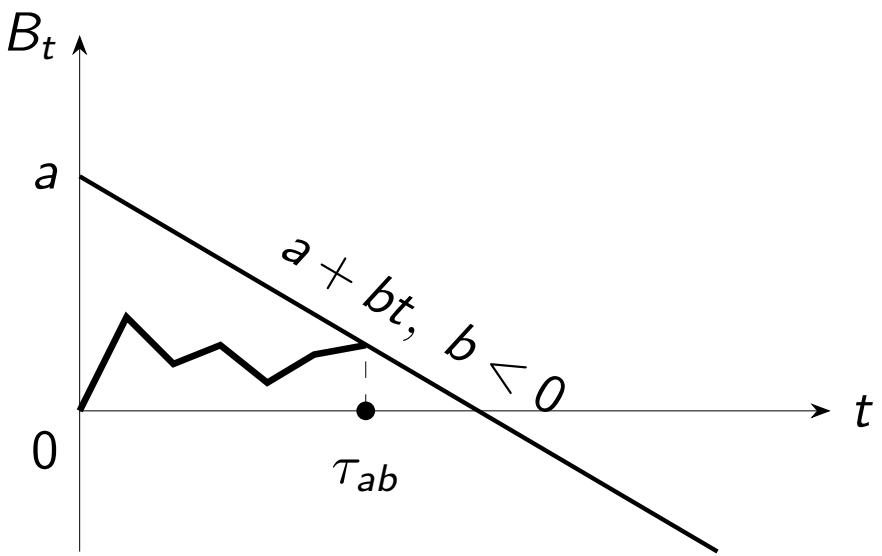
Probability density function of σ_d is

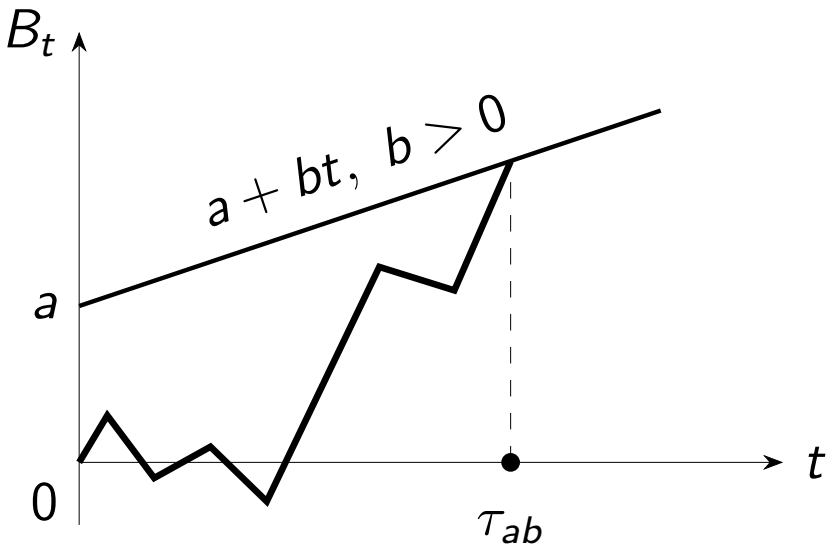
$$p_{\sigma_d} = 2\sum_{n=0}^{\infty}(-1)^n \frac{a(2n+1)}{t}\phi_t(a(2n+1)) = 2\sum_{n=0}^{\infty}(-1)^n p_{\sigma_{a(2n+1)}}(t).$$

Cumulative density function of σ_d is

$$P(\sigma_d \leq t) = \int_0^t p_{\sigma_d}(s)ds = 4\sum_{n=0}^{\infty}(-1)^n \left[1 - \Phi\left(\frac{a(2n+1)}{\sqrt{t}}\right)\right].$$

Properties of $\tau_{ab} = \inf\{t \geq 0 : B_t - bt \geq a\}, a > 0.$





Girsanov theorem and its corollary

Let $(\Omega, \mathcal{F}, F, \mathrm{P})$ be a filtered probability space. Measure P_T^λ is such that $d\mathrm{P}_T^\lambda = e^{\lambda B_T - \frac{\lambda^2}{2} T} dP_T$.

Girsanov theorem

With respect to measure P_T^λ a process with drift $B_t^\lambda = B_t - \lambda t,\ t \leq T$ is a standard Brownian motion

$$\mathrm{Law}(B_t^\lambda, t \leq T | \mathrm{P}_T^\lambda) = \mathrm{Law}(B_t, t \leq T | \mathrm{P}_T).$$

Corollary

If $G_T(x) = G_T(x_t, t \leq T)$ is a measurable functional on the space of countinious functions $x = (x_t)_{t \leq T}$, then for non-negative or bounded functionals

$$\mathrm{Law}(G_T(B^\lambda) | \mathrm{P}_T^\lambda) = \mathrm{Law}(G_T(B) | \mathrm{P}_T),$$

and therefore

$$\mathrm{E} G_T(B^\lambda) = \mathrm{E} e^{-\lambda B_T - \frac{\lambda^2}{2} T} G_T(B).$$

Girsanov theorem and its corollary for stopping times

Let τ be such that $\mathrm{E}Z_\tau^\lambda = 1$, where $Z_t^\lambda = e^{\left(\lambda B_{t\wedge\tau} - \frac{\lambda^2}{2}(t\wedge\tau)\right)}$, $t \geq 0$. Measure $\tilde{\mathrm{P}}^\lambda$ is such that $d\tilde{\mathrm{P}}^\lambda = Z_\tau^\lambda dP$.

Girsanov theorem

With respect to measure $\tilde{\mathrm{P}}^\lambda$ a process with drift $\tilde{B}_t^\lambda = B_t - \lambda(t \wedge \tau)$ is a standard Brownian motion

$$\mathrm{Law}(\tilde{B}^\lambda|\tilde{\mathrm{P}}^\lambda) = \mathrm{Law}(B|\mathrm{P}).$$

Corollary

If τ is such that $\mathrm{E}e^{\lambda B_\tau - \frac{\lambda^2}{2}\tau} = 1$, then

$$\mathrm{E}_{\tilde{\mathrm{P}}^\lambda}(G_\tau(\tilde{B}^\lambda)\mathrm{I}(\tau < \infty)) = \mathrm{E}(G_\tau(B)\mathrm{I}(\tau < \infty)),$$

and therefore

$$\mathrm{E}[G_\tau(\tilde{B}^\lambda)\mathrm{I}(\tau < \infty)] = \mathrm{E}\left[e^{-\lambda B_\tau - \frac{\lambda^2}{2}\tau} G_\tau(B)\mathrm{I}(\tau < \infty)\right].$$

Properties of $\tau_{ab} = \inf\{t \geq 0 : B_t - bt \geq a\}, a > 0.$

Laplace transform is

$$\mathrm{E}e^{-\lambda\tau_{ab}} = e^{-a(b+\sqrt{b^2+2\lambda})}.$$

Probability density function of τ_{ab} is

$$p_{\tau_{ab}} = \frac{a}{t}\phi_t(a+bt) = \frac{1}{\sqrt{2\pi t}}\frac{a}{t}e^{\frac{-(a+bt)^2}{2t}}.$$

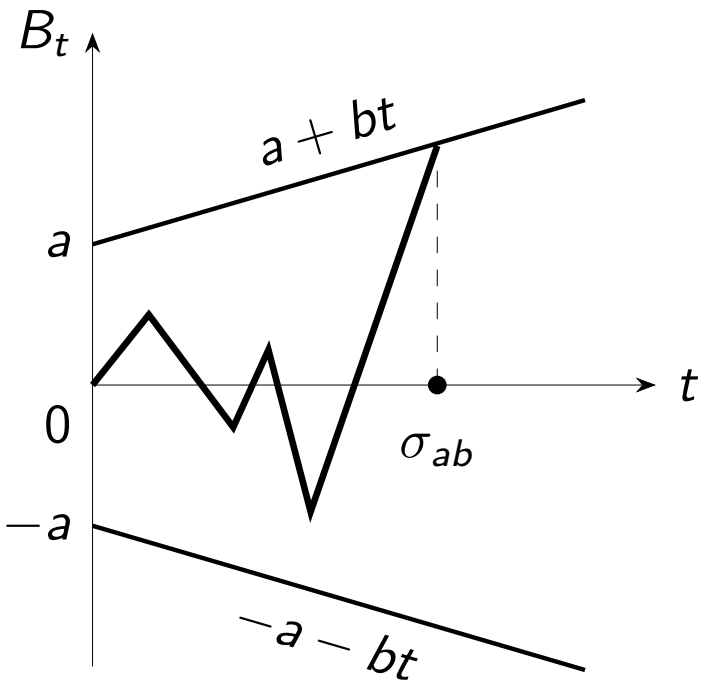
Cumulative density function of τ_{ab} is

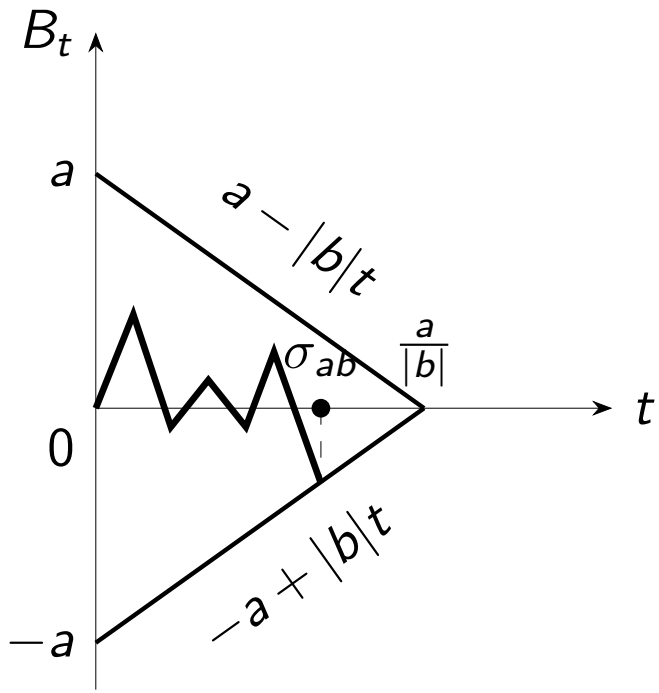
$$\mathrm{P}(\tau_{ab} \leq t) = \int_0^t \frac{a}{s}\phi_t(a+bs)ds = 1-\Phi\left(\frac{a+bt}{\sqrt{t}}\right)+e^{-2ab}\Phi\left(\frac{bt-a}{\sqrt{t}}\right).$$

Expectation of τ_{ab} is

$$\mathrm{E}\tau_{ab} = \begin{cases} \frac{a}{|b|}, & b < 0 \\ \infty, & b > 0 \end{cases}.$$

Properties of $\sigma_{ab} = \inf\{t \geq 0 : |B_t| \geq a + bt\}$, $a > 0$





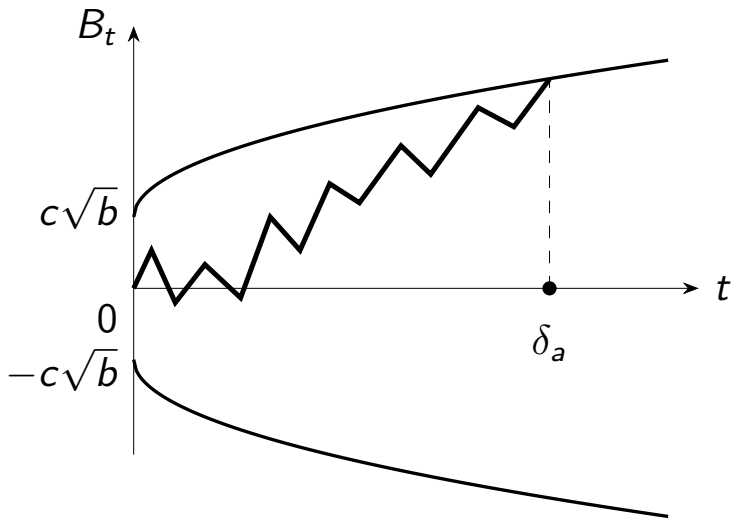
Properties of $\sigma_{ab} = \inf\{t \geq 0 : |B_t| \geq a + bt\}$, $a > 0$

Cumulative density function of σ_{ab} is

$$P(\tau \leq t) = 1 - \mathrm{P}(\tau > t) = 1 - \sum_{k=-\infty}^{+\infty} e^{-8abk^2} \frac{1}{\sqrt{2\pi t}} \int_{-a-bt}^{a+bt} e^{-(x-4ak)^2/(2t)} dt +$$

$$+ \sum_{k=-\infty}^{+\infty} e^{-2ab(2k+1)^2} \frac{1}{\sqrt{2\pi t}} \int_{-a-bt}^{a+bt} e^{-(x-2a(2k+1))^2/(2t)} dt.$$

Properties of $\delta_a = \inf\left\{t \geq 0 : |B_t| \geq c\sqrt{t+b}\right\}$



Properties of $\delta_a = \inf\left\{t \geq 0 : |B_t| \geq c\sqrt{t+b}\right\}$

Second Wald's identity for Brownian motion

For random stopping time τ (with respect to $(\mathcal{F}_t^B)_{t\geq 0}$) with $\mathbb{E}\tau < \infty$

$$\mathrm{E}B_\tau^2 = \mathrm{E}\tau.$$

Expectation of δ_a is

$$\mathrm{E}B_{\delta_a} = \frac{c^2 b}{1-c^2}.$$

Expectation of $\delta_b = \inf\left\{t \leq b : |B_t| \geq c\sqrt{b-t}\right\}$ is

$$\mathrm{E}\delta_b = \frac{c^2 b}{1+c^2}.$$

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